

Experiment 5.3: Design and Analysis of an FIR High-Pass Filter Using the Hanning Window Technique in MATLAB

Aim: To design and analyze an FIR High-Pass Filter using the Hanning Window technique in MATLAB.

MATLAB Code

```
clc;
clear;
close all;

%% Save Command Window Output
diary('Exp5_3_Output.txt');

%% --- Input Parameters ---
fp = 1000;      % High-pass cutoff frequency (Hz)
fs = 8000;      % Sampling frequency (Hz)
N = 50;         % Filter order

wc = fp/(fs/2); % Normalized cutoff (0-1)

%% --- FIR High Pass using Hanning Window ---
b_hann = fir1(N, wc, 'high', hanning(N+1));

disp('--- High-Pass Coefficients (Hanning Window) ---');
disp(b_hann);

%% --- Frequency Response Plot ---
figure(1);
freqz(b_hann,1);
title('High-pass FIR using Hanning Window');

%% Stop Saving Command Window
diary off;

%% Save Figure (300 DPI)
exportgraphics(figure(1), 'Exp5_3_Hanning_Window.png', 'Resolution', 300);

%% Completion Message
disp(' ');
disp('=====');
disp('Experiment 5.3 Completed Successfully');
disp('Files Saved:');
disp('Exp5_3_Output.txt');
disp('Exp5_3_Hanning_Window.png');
disp('=====');
```

Output

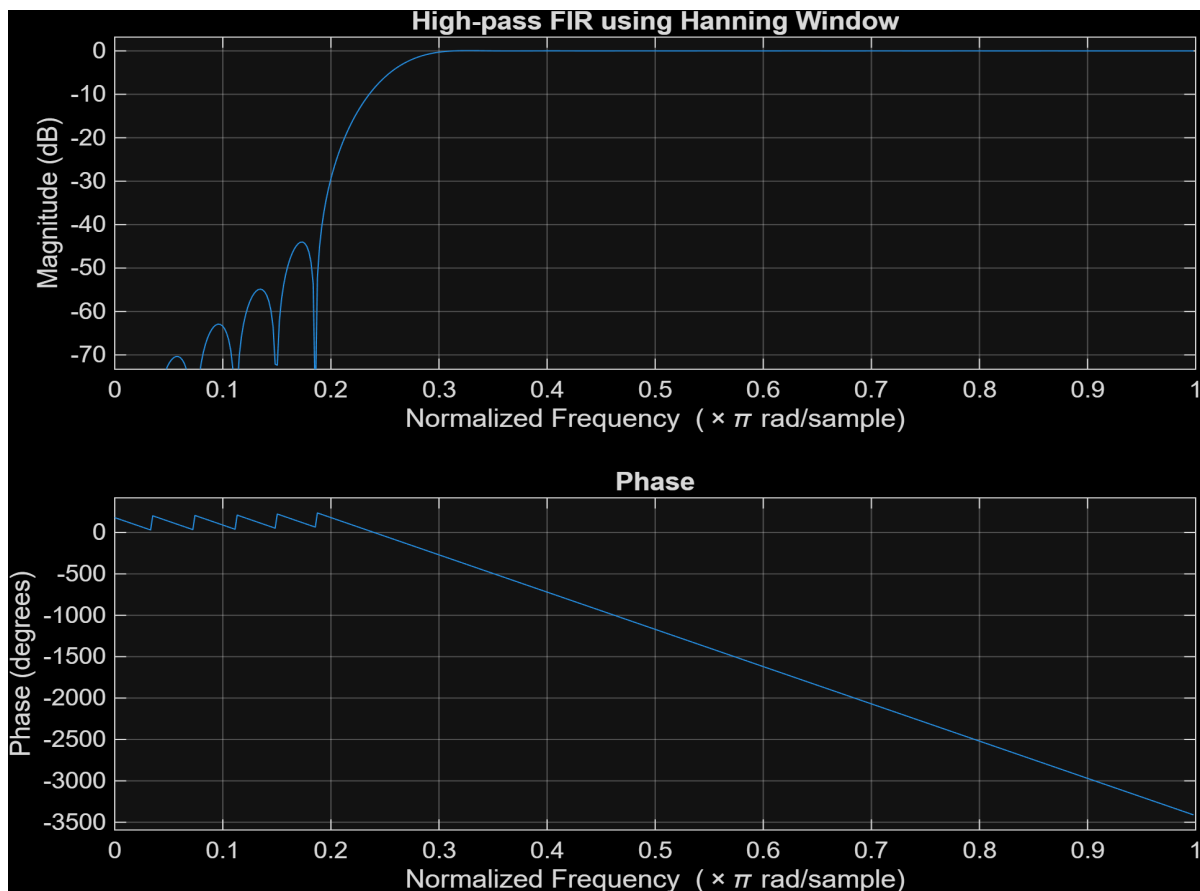
```
--- High-Pass Coefficients (Hanning Window) ---
-0.0000
  0
  0.0003
  0.0008
  0.0009
  0
-0.0020
-0.0038
-0.0035
  0
  0.0057
  0.0100
  0.0087
  0
-0.0127
-0.0216
-0.0183
```

```

0
0.0267
0.0464
0.0410
0
-0.0726
-0.1568
-0.2243
0.7500
-0.2243
-0.1568
-0.0726
0
0.0410
0.0464
0.0267
0
-0.0183
-0.0216
-0.0127
0
0.0087
0.0100
0.0057
0
-0.0035
-0.0038
-0.0020
0
0.0009
0.0008
0.0003
0
-0.0000

```

Output Image



Result: The FIR High-Pass filter using the Hanning Window technique was successfully designed and analyzed in MATLAB. The frequency response and phase response were obtained successfully.